CLIENT	**:**	**GMR**
PROJECT	**:**	**Installation of door for hanger at Hyderabad Airport**
LOCATION	**:**	**Hyderabad**
PROJECT NO.	**:**	**1729**
DOCUMENT NO.	**:**	**STATIC-REPORT-001**

0	05.11.2022	For Information	R.H.B	H.A.B	H.A.B
Rev No.	**Date**	**Description**	**Prepared By**	**Checked By**	**Approved By**

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1. Introduction purpose and scope

Hyderabad International Airport is located in Hyderabad, Telangana state of India. The airport has three hangers for maintenance of planes adjacent to each other viz. 'Hanger 1', 'Hanger 2' & 'Hanger 3'. Hanger 3 has a sliding door installed. Purpose of the door is to achieve weather protection from all sides in working area. Hanger 1 & 2 currently has no door and concerned authority wants to install new doors. NIHVA Technologies has been appointed to install the doors.

Purpose of this document to establish design and assumptions structural framing system to support the door. Expansion of Hanger 1 & 2 is to be done 4.75m from existing structure for which roof truss frame is to be designed on which the door should be installed. In this calculation it will be established that new roof truss frame can take additional dead load and operating load due to the door. Also, wind load will be applied on the new truss frame which in return will transfer to MWFRS of existing building. Additional frame is required be designed to install the door. This framing will be connected to existing truss through bolted connections.

2. Design Philosophy

As door is installed on new framing, various DL and operating loads acting at different points on frame are transferred to existing truss. By truss action, the loads are then transferred to nearest columns and then the load is dispersed to foundation. The lateral loads due to wind shall be acting when door is closed. Wind braces are provided at locations mentioned in drawing. Lateral load is partially transferred through mullion to its foundation and partially to wind braces. Lateral load in braces is transferred through main wind force resisting system of roof truss. The end frames of structure are braced, which in turn will transfer load to respective foundations. The complete arrangement is modeled as 3D frame and analyzed/designed for worst combination of loads through STAAD.pro. Input and output files are enclosed which are self-explanatory. From the STAAD output, the support reactions for various primary load cases are tabulated accordingly.

3. Design Codes and Standards

Design and fabrication of equipment will be carried out as per following codes and standards.

Code / Standard	Applicability	Remark
IS800-2007	Design of steel structures	--
ASME Section IX	Welding & Brazing Qualifications	--
IS 875-Part I, II, III	For all load application and load cases other than earthquake load	--
IS1893	For seismic load applications	--

4. Material Specification For Construction

Material Specifications				
Sr. No.	**Materials**		**Specification**	**Minimum Strength**
1	Hot Rolled	Tubes	Indian Standards	Fy = 310 N/mm^2
		Channels	Indian Standards	Fy = 250 N/mm^2
		Beams, Columns, Plates, Angles, Bars	Indian Standards	Fy = 250 N/mm^2
2	Cold Formed	Handrail Pipe	Indian Standards	Fy = 230 N/mm^2
3	High Strength Bolts		Indian Standards	Fy = 630 N/mm^2
4	Nuts		Indian Standards	-
5	Washers		Indian Standards	-
6	Taper Washers		Indian Standards	-
7	Grating (Hot Dip Galvanized)		Bearing Bar 25x5 @ 30 C/C & 6 mm square twisted bar @ 100 mm C/C	-

5. Load Calculations

5.1 Dead Load

Self-weight of various structural members is applied using “self-weight” command in STAAD Pro.
Dead load is applied for roof sheeting on existing roof truss
From the received input spacing of new truss frame and adjacent truss is 5m and assuming dead load of sheeting as 0.2kN/sqm
DL on truss = 0.2 x 4.75/2 = 0.4875 kN/m

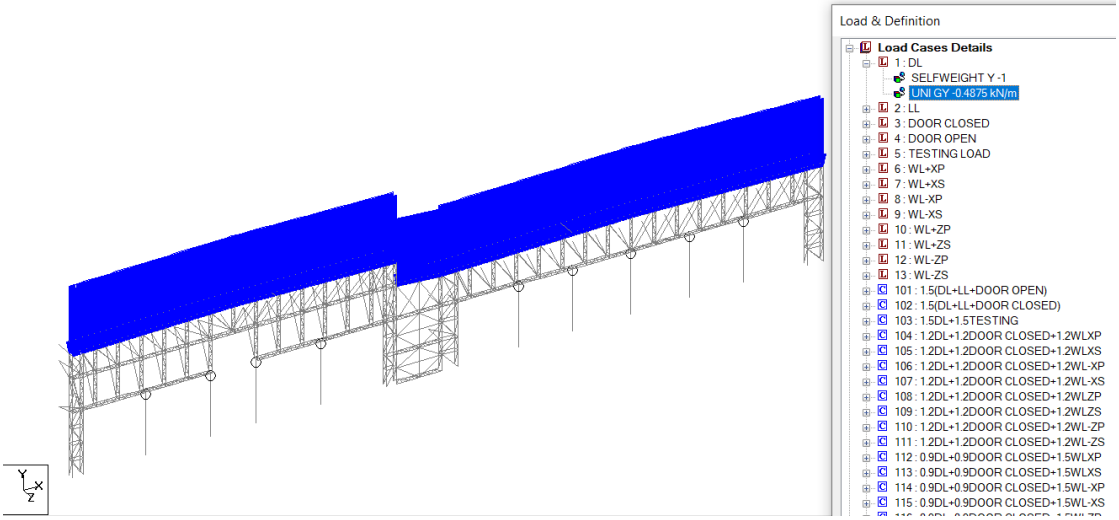


Fig – 1 (Dead load 0.4875kN/m applied in staad pro model as UDL)

5.2 Live load

Live load on top of existing truss is applied in live load case.
Assuming live load of 0.75kN/m

LL on truss = 0.75 x 4.75/2 = 1.78 kN/m

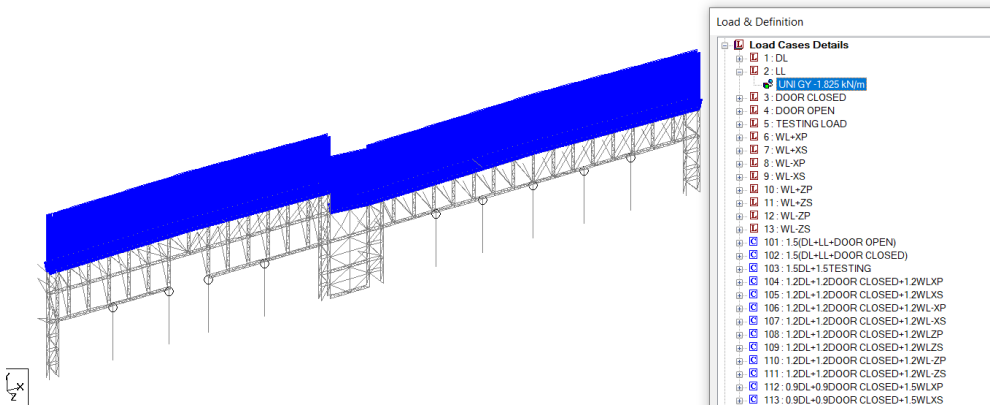


Fig – 2 (Dead load 1.78kN/m applied in staad pro model as UDL)

5.3 Equipment Loads

a. Door closed case

All dead loads for door are provided by ASSA ABLOY in drawing number 77446 & 77450
Other dead loads of the door are applied in 2 cases

Case 1: Door closed
For 84m Hanger door -
Distributed load per drawing = 33.3kN – distributed over 16.9m length = 33.3/16.9 = 1.97kN/m
Distributed load per drawing = 45.3kN – distributed over 12m length = 45.3/12 = 3.775kN/m
Applied as UDL in Staad pro model '1729 HYD Hanger door 05-11-2022'

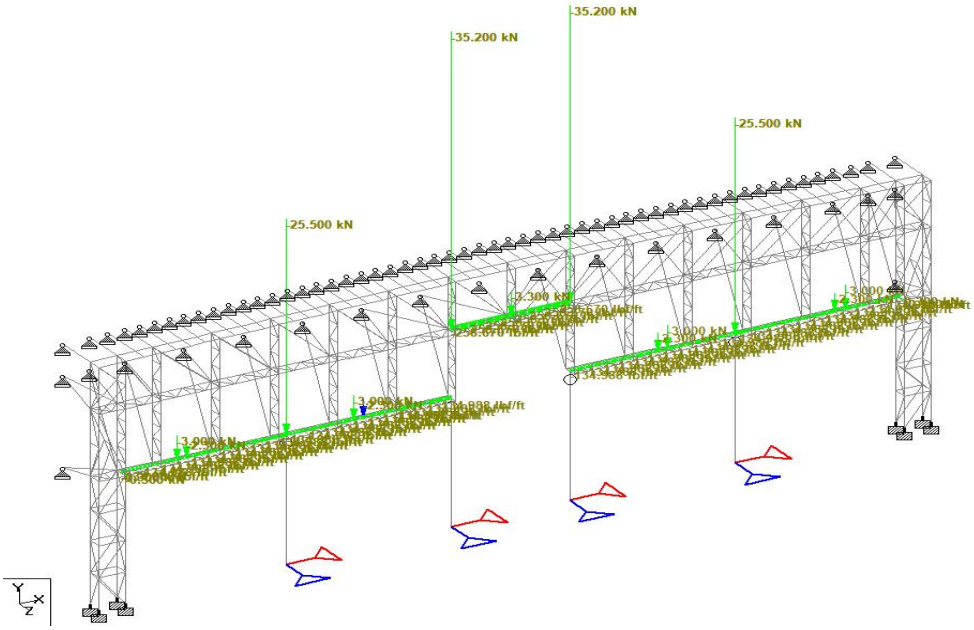


Fig – 3 (All loads for door closed per dwg no 77446 applied as point load in staad pro model)

For 99m Hanger door -
Distributed load per drawing = 33kN – distributed over 15.34m length = 33/15.34 = 2.15kN/m
Applied as UDL in Staad pro model '1729 HYD Hanger door 05-11-2022'

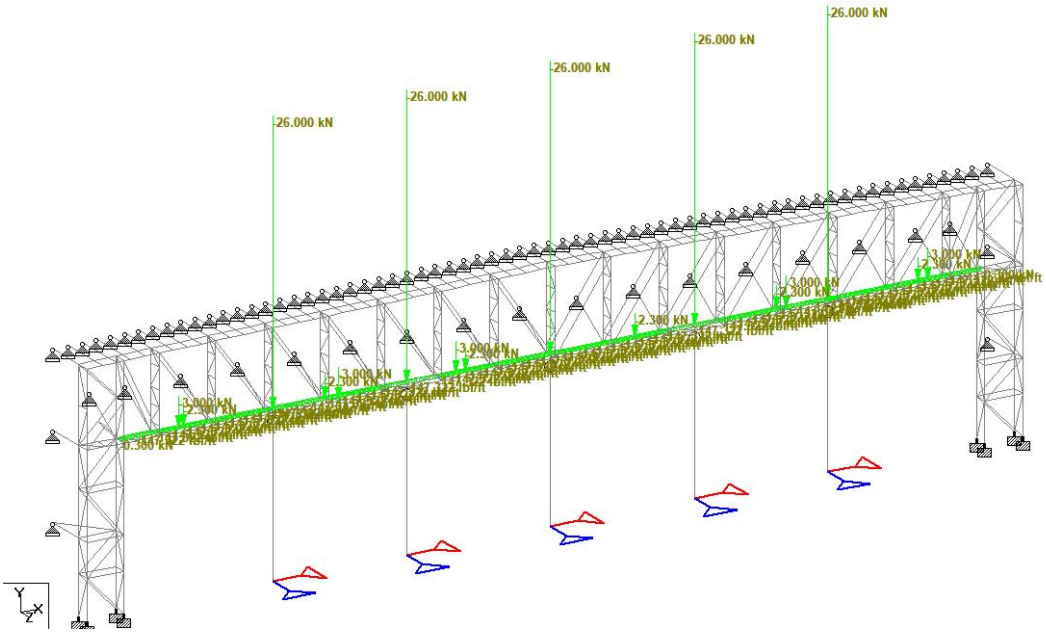


Fig – 4 (All loads for door closed per dwg no 77450 applied as point load in staad pro model)

b. *Door open case (Operating load)*

For 84m Hanger door -
Distributed load per drawing = 1.1kN – distributed over 16.9m length = 1.1/16.9 = 0.065kN/m
Distributed load per drawing = 1.1kN – distributed over 16.9m length = 1.1/12 = 0.091kN/m
Applied as UDL in Staad pro model '1729 HYD Hanger door 05-11-2022'

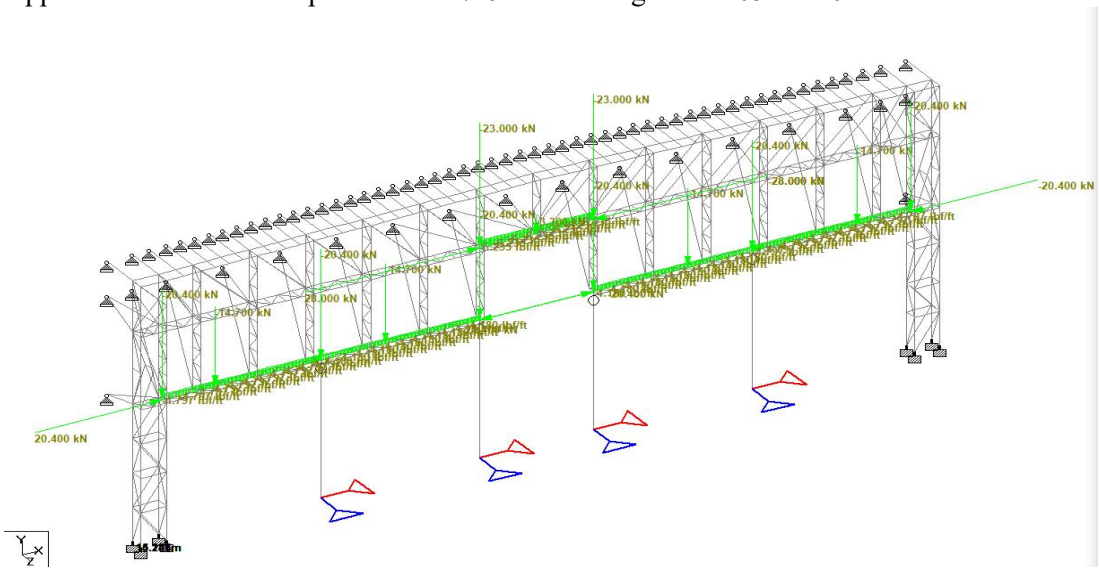


Fig – 5 (All loads for door open per dwg no 77446 applied as point load in staad pro model)

For 99m Hanger door –
Distributed load per drawing = 1.1kN – distributed over 15.75m length = 1.1/15.71 = 0.07kN/m
Applied as UDL in Staad pro model '1729 HYD Hanger door 05-11-2022'

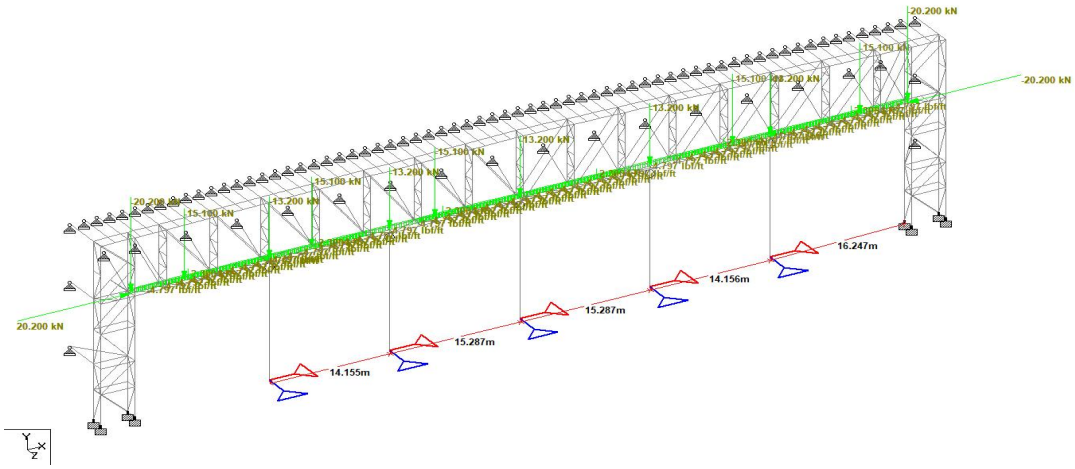


Fig – 6 (All loads for door open per dwg no 77450 applied as point load in staad pro model)

c. *Testing load*

Additional load to be applied during testing procedure is applied in “testing” load case.

For 84m Hanger door -

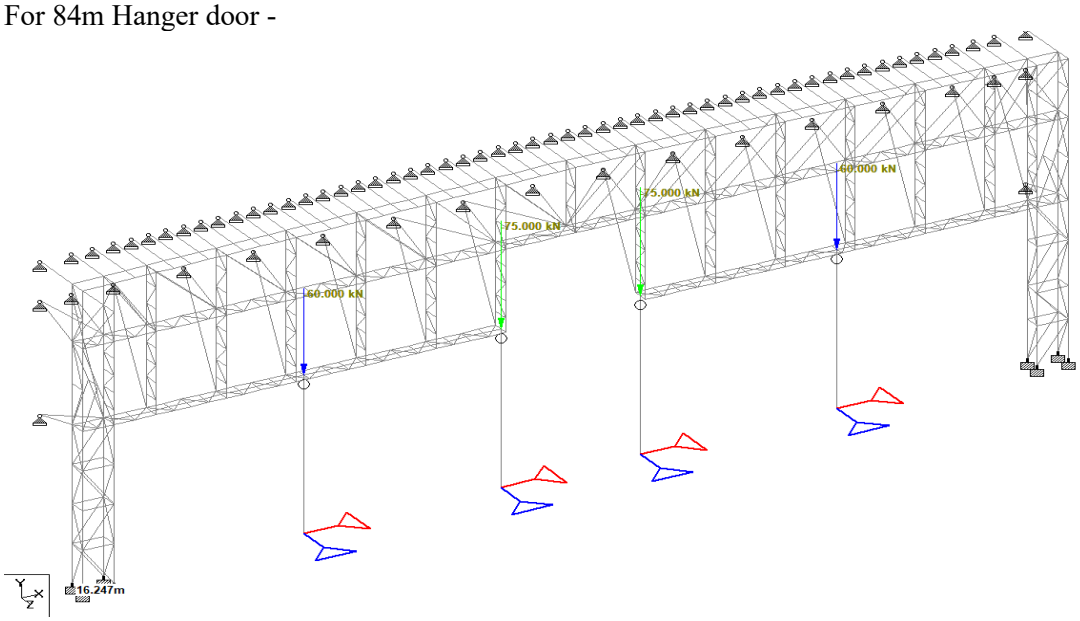


Fig – 7 (All loads for testing per dwg no 77446 applied as point load in staad pro model)

For 99m Hanger door –

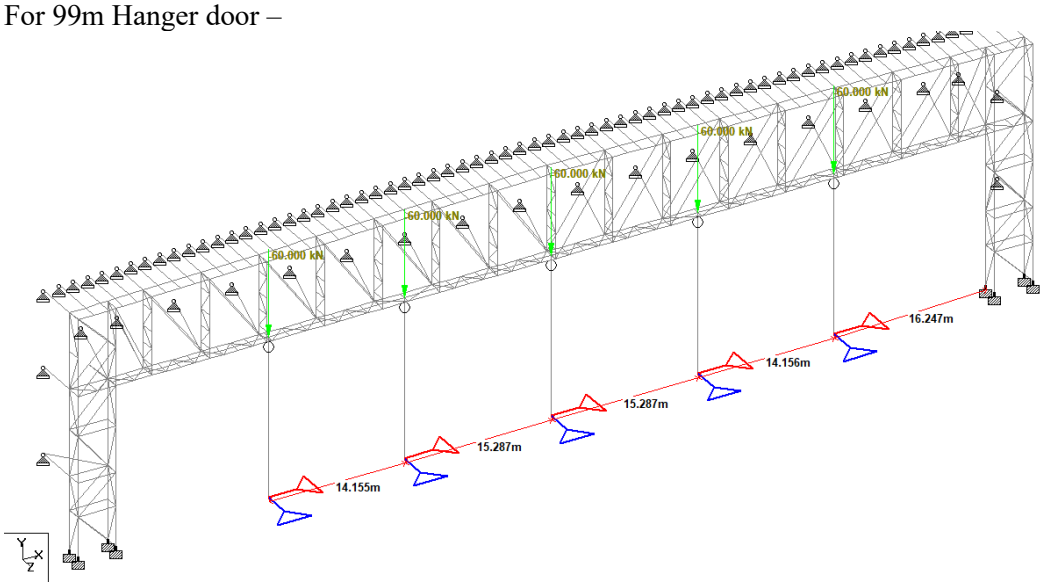


Fig – 8 (All loads for testing per dwg no 77450 applied as point load in staad pro model)

5.4 Wind load

Wind load shall be acting on the mullions on which door sheeting & frame on which purlins will be supported and will be transferred to new truss frame.

For 84m hanger door –

Wind Load Calculation				
City - Hyderabad	**Basic Wind Speed As per Apendix A (IS-875 Part 3)**			
Hyderabad	44	m/s		
K1=	1	As per IS 875 Part III (table-1)		
K2=	1.12	As per IS 875 Part III (table-2)		
K3=	1	As per IS 875 Part III (table-2)		
K4=	1.15			
Vz=	Vb*K1*K2*K3*K4			
Vz=	56.67	m/s	As per IS 875	
			Clause No.5.3.3.1	
Pz=	$0.6(Vz)^2$			
Pz=	1.9270	kN/m2		
Pd1=0.7Pz	1.35	kN/m2	=PD1	
ka=	0.80			
kd=	0.9			
kc=	0.9			
	1.25		=Pd2	
	Since Pd2<Pd1, we will use Pd1			
Height building Ratio	H/w	0.39		
Building Plan ratio	L/w	1.18		

For Cladding -

Wind direction	Load case	Ext. Pressure coef. Cpe				Cpi
		A	B	C	D	
Wind in +X direction	+X(P)	0.7	-0.2	-0.5	-0.5	0.5
	+X(S)	0.7	-0.2	-0.5	-0.5	-0.5
Wind in -X direction	-X(P)	-0.2	0.7	-0.5	-0.5	0.5
	-X(S)	-0.2	0.7	-0.5	-0.5	-0.5
Wind in +Z direction	+Z(P)	-0.5	-0.5	0.7	-0.2	0.5
	+Z(S)	-0.5	-0.5	0.7	-0.2	-0.5
Wind in -Z direction	-Z(P)	-0.5	-0.5	-0.2	0.7	0.5
	-Z(S)	-0.5	-0.5	-0.2	0.7	-0.5

$C_p = C_{pe}-C_{pi}$				Pressure	Pressure on each face kN/m^2			
A	B	C	D	Pz	A	B	C	D
0.2	-0.7	-1	-1	1.35	0.27	-0.94	-1.35	-1.35
1.2	0.3	0	0	1.35	1.62	0.40	0.00	0.00
-0.7	0.2	-1	-1	1.35	-0.94	0.27	-1.35	-1.35
0.3	1.2	0	0	1.35	0.40	1.62	0.00	0.00
-1	-1	0.2	-0.7	1.35	-1.35	-1.35	0.27	-0.94
0	0	1.2	0.3	1.35	0.00	0.00	1.62	0.40
-1	-1	-0.7	0.2	1.35	-1.35	-1.35	-0.94	0.27
0	0	0.3	1.2	1.35	0.00	0.00	0.40	1.62

For roof –

Wind direction	Load case	Ext. Pressure coef. Cpe				Cpi
		E	F	G	H	
Wind in +X direction	+X(P)	-0.9	-0.9	-0.4	-0.4	0.5
	+X(S)	-0.9	-0.9	-0.4	-0.4	-0.5
Wind in -X direction	-X(P)	-0.4	-0.4	-0.9	-0.9	0.5
	-X(S)	-0.4	-0.4	-0.9	-0.9	-0.5
Wind in +Z direction	+Z(P)	-0.8	-0.4	-0.8	-0.4	0.5
	+Z(S)	-0.8	-0.4	-0.8	-0.4	-0.5
Wind in -Z direction	-Z(P)	-0.8	-0.4	-0.8	-0.4	0.5
	-Z(S)	-0.8	-0.4	-0.8	-0.4	-0.5

$C_p = C_{pe}-C_{pi}$				Pressure	Pressure on each face kN/m^2			
E	F	G	H	Pz	E	F	G	H
-1.4	-1.4	-0.9	-0.9	1.35	-1.89	-1.89	-1.21	-1.21
-0.4	-0.4	0.1	0.1	1.35	-0.54	-0.54	0.13	0.13
-0.9	-0.9	-1.4	-1.4	1.35	-1.21	-1.21	-1.89	-1.89
0.1	0.1	-0.4	-0.4	1.35	0.13	0.13	-0.54	-0.54
-1.3	-0.9	-1.3	-0.9	1.35	-1.75	-1.21	-1.75	-1.21
-0.3	0.1	-0.3	0.1	1.35	-0.40	0.13	-0.40	0.13
-1.3	-0.9	-1.3	-0.9	1.35	-1.75	-1.21	-1.75	-1.21
-0.3	0.1	-0.3	0.1	1.35	-0.40	0.13	-0.40	0.13

For 99m Hanger door -

Wind Load Calculation				
City - Hyderabad	**Basic Wind Speed As per Apendix A (IS-875 Part 3)**			
Hyderabad	44	m/s		
K1=	1	As per IS 875 Part III (table-1)		
K2=	1.12	As per IS 875 Part III (table-2)		
K3=	1	As per IS 875 Part III (table-2)		
K4=	1.15			
Vz=	Vb*K1*K2*K3*K4			
Vz=	56.67	m/s	As per IS 875 Clause No.5.3.3.1	
Pz=	$0.6(Vz)^2$			
Pz=	1.9270	kN/m2		
Pd1=0.7Pz	1.35	kN/m2	=PD1	
ka=	0.80			
kd=	0.9			
kc=	0.9			
	1.25		=Pd2	
	Since Pd2<Pd1, we will use Pd1			
Height building Ratio	H/w	0.26		
Building Plan ratio	L/w	1.01		

For Cladding –

Wind direction	Load case	Ext. Pressure coef. Cpe				Cpi
		A	B	C	D	
Wind in +X direction	+X(P)	0.7	-0.2	-0.5	-0.5	0.5
	+X(S)	0.7	-0.2	-0.5	-0.5	-0.5
Wind in -X direction	-X(P)	-0.2	0.7	-0.5	-0.5	0.5
	-X(S)	-0.2	0.7	-0.5	-0.5	-0.5
Wind in +Z direction	+Z(P)	-0.5	-0.5	0.7	-0.2	0.5
	+Z(S)	-0.5	-0.5	0.7	-0.2	-0.5
Wind in -Z direction	-Z(P)	-0.5	-0.5	-0.2	0.7	0.5
	-Z(S)	-0.5	-0.5	-0.2	0.7	-0.5

$C_p = C_{pe}-C_{pi}$				Pressure	Pressure on each face kN/m^2			
A	B	C	D	Pz	A	B	C	D
0.2	-0.7	-1	-1	1.35	0.27	-0.94	-1.35	-1.35
1.2	0.3	0	0	1.35	1.62	0.40	0.00	0.00
-0.7	0.2	-1	-1	1.35	-0.94	0.27	-1.35	-1.35
0.3	1.2	0	0	1.35	0.40	1.62	0.00	0.00
-1	-1	0.2	-0.7	1.35	-1.35	-1.35	0.27	-0.94
0	0	1.2	0.3	1.35	0.00	0.00	1.62	0.40
-1	-1	-0.7	0.2	1.35	-1.35	-1.35	-0.94	0.27
0	0	0.3	1.2	1.35	0.00	0.00	0.40	1.62

For roof –

Wind direction	Load case	Ext. Pressure coef. Cpe				Cpi
		E	F	G	H	
Wind in +X direction	+X(P)	-0.9	-0.9	-0.4	-0.4	0.5
	+X(S)	-0.9	-0.9	-0.4	-0.4	-0.5
Wind in -X direction	-X(P)	-0.4	-0.4	-0.9	-0.9	0.5
	-X(S)	-0.4	-0.4	-0.9	-0.9	-0.5
Wind in +Z direction	+Z(P)	-0.8	-0.4	-0.8	-0.4	0.5
	+Z(S)	-0.8	-0.4	-0.8	-0.4	-0.5
Wind in -Z direction	-Z(P)	-0.8	-0.4	-0.8	-0.4	0.5
	-Z(S)	-0.8	-0.4	-0.8	-0.4	-0.5

$C_p = C_{pe}-C_{pi}$				Pressure	Pressure on each face kN/m^2			
E	F	G	H	Pz	E	F	G	H
-1.4	-1.4	-0.9	-0.9	1.35	-1.89	-1.89	-1.21	-1.21
-0.4	-0.4	0.1	0.1	1.35	-0.54	-0.54	0.13	0.13
-0.9	-0.9	-1.4	-1.4	1.35	-1.21	-1.21	-1.89	-1.89
0.1	0.1	-0.4	-0.4	1.35	0.13	0.13	-0.54	-0.54
-1.3	-0.9	-1.3	-0.9	1.35	-1.75	-1.21	-1.75	-1.21
-0.3	0.1	-0.3	0.1	1.35	-0.40	0.13	-0.40	0.13
-1.3	-0.9	-1.3	-0.9	1.35	-1.75	-1.21	-1.75	-1.21
-0.3	0.1	-0.3	0.1	1.35	-0.40	0.13	-0.40	0.13

6. *Primary Loads and Load Combinations*

Primary loads

- Load 1 Dead Load (DL)
- Load 2 Live Load (LL)
- Load 3 Door Open
- Load 4 Door Closed
- Load 5 Testing load
- Load 6 Wind load XP
- Load 7 Wind load XS
- Load 8 Wind load -XP
- Load 9 Wind load -XS
- Load 10 Wind load ZP
- Load 11 Wind load ZS
- Load 12 Wind load -ZP
- Load 13 Wind load -ZS

Load combinations

- LOAD COMB 101 1.5DL+1.5LL+1.5DOOR OPEN
- LOAD COMB 102 1.5DL+1.5LL+1.5DOOR CLOSED
- LOAD COMB 103 1.5DL+1.5TESTING
- LOAD COMB 104 1.2DL+1.2DOOR CLOSED+1.2WLXP
- LOAD COMB 105 1.2DL+1.2DOOR CLOSED+1.2WLXS
- LOAD COMB 106 1.2DL+1.2DOOR CLOSED+1.2WL-XP
- LOAD COMB 107 1.2DL+1.2DOOR CLOSED+1.2WL-XS
- LOAD COMB 108 1.2DL+1.2DOOR CLOSED+1.2WLZP
- LOAD COMB 109 1.2DL+1.2DOOR CLOSED+1.2WLZS
- LOAD COMB 110 1.2DL+1.2DOOR CLOSED+1.2WL-ZP
- LOAD COMB 111 1.2DL+1.2DOOR CLOSED+1.2WL-ZS
- LOAD COMB 112 0.9DL+0.9DOOR CLOSED+1.5WLXP
- LOAD COMB 113 0.9DL+0.9DOOR CLOSED+1.5WLXS
- LOAD COMB 114 0.9DL+0.9DOOR CLOSED+1.5WL-XP
- LOAD COMB 115 0.9DL+0.9DOOR CLOSED+1.5WL-XS
- LOAD COMB 116 0.9DL+0.9DOOR CLOSED+1.5WLZP

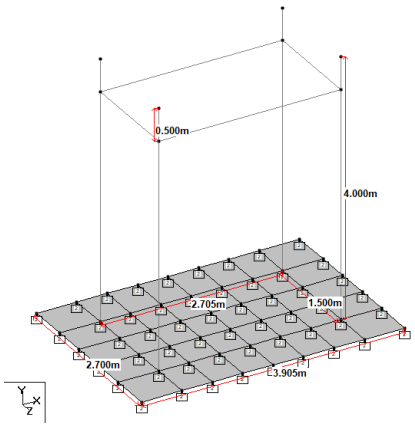
- LOAD COMB 117 0.9DL+0.9DOOR CLOSED+1.5WLZS
- LOAD COMB 118 0.9DL+0.9DOOR CLOSED+1.5WL-ZP
- LOAD COMB 119 0.9DL+0.9DOOR CLOSED+1.5WL-ZS
- LOAD COMB 201 DL+LL+DOOR OPEN
- LOAD COMB 201 DL+LL+DOOR OPEN
-

SALT Design Services / NIHVA	**Megadoor installation at Hyderabad**		**Computed By:**	Rishabh Bhirud
	Client:	GMR	**Date;**	05.Nov.2022
	Project:	1729	**Verified By:**	Harshal Bhandari
	Title:	HYD HANGER DOOR	**Date:**	05.Nov.2022
		Hydrabad (Footing design)	**Page:**	

Design philosophy

As door is installed on new framing, various loads acting at different points on frame are transferred to the truss. By truss action, the loads are then transferred to nearest columns and then the is dispersed to foundation. So we have modelled each footing group in staad as plate element. Pedestals and tie beams are modelled as line elements. Superstructure reactions are taken from "1729 HYD hanger door 05-11-2022" staad model. The loads are applied on top of pedestal for each load case. Limit state and working stress load combinations are applied for strength and serviceability design. Modolous of subgrade is calculated from soil report.

For footing group 1 (84m)



Calculation of steel required for footing raft :-

Thickness of footing =	550	mm	
Clear cover =	50	mm	
Dia of bar =	12	mm	
Spacing of bar =	140	mm	
Grade of concrete =	M	30	
Grade of steel =	Fe	500	
Bending moment=	159	kN-m	Max Mx+Mxy (From staad model) Load comb:
Effective depth =	506	mm	

$$A_s = \frac{0.5\, f_c}{f_y}\left(1 - \sqrt{1 - \frac{4.6\, M_u}{f_c b\, d^2}}\right) bd$$

SALT Design Services / NIHVA	Megadoor installation at Hyderabad		Computed By:	Rishabh Bhirud
	Client:	GMR	Date;	05.Nov.2022
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	Title:	HYD HANGER DOOR	Date:	05.Nov.2022
		Hydrabad (Footing design)	Page:	

A_{st} = 740.8033 mm^2

$A_{st\ min}$ = 660 mm^2

Hence Ast req= 740.8033 mm^2

A_{st} provided = 807.8381 mm^2 >Ast reqd, Hence OK

Hence provide 12 mm dia steel at 140c/c top and bottom steel

Check for uplift :-

Uplift load = -267 kN Refer reaction sheet for footing group 1(84m)

Total downward load = ∑ Fy From L/C 204 "footing group 1 (84m) staad)

= 805.577 kN

Total uplift load < total downward load, Hence safe

Check for Sliding :-

Resultant force= SRSS of Max of Fx + Fz

= 158 kN Refer reaction sheet for footing group 1(84m)

Coefficient of friction used = 0.57

Total downward force = ∑ Fy From L/C 211 "footing group 1 (84m) staad)

= 490 kN

= 279.3 kN Hence safe

Check for Overturning :-

Consider load combination 219. Maximum moment on the footing due to wind load in Z direction is 93kN-m

Hence, P/A±M/Z =

P = 1330kN (Summation staad reactions for 219L/C)

A = 10.53

M= 93kN-m (From Staad reaction for L/C219)

Z = 4.7385 m3

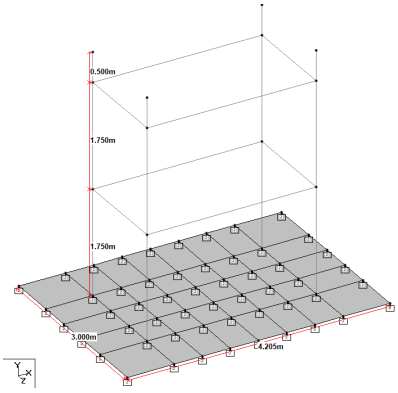
Hence, P/A+M/Z = 145.9675 kN/m2 < SBC 13x1.25 T/m2

Hence, P/A-M/Z = 106.6441 kN/m2 > 0T/m2

As both reactions are positive overturning wont happen

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	Title:	HYD HANGER DOOR	**Date:**	05.Nov.2022
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For footing group 2 (84m)



Calculation of steel required for footing raft :-

Thickness of footing =	600	mm	
Clear cover =	50	mm	
Dia of bar =	12	mm	
Spacing of bar =	150	mm	
Grade of concrete =	M	30	
Grade of steel =	Fe	500	
Bending moment=	125	kN-m	Max Mx+Mxy (From staad model)
Effective depth =	556	mm	

$$A_s = \frac{0.5\, f_c}{f_y}\left(1 - \sqrt{1 - \frac{4.6\, M_u}{f_c b\, d^2}}\right) bd$$

A_{st} =	525.3598	mm^2
$A_{st\ min}$ =	720	mm^2
Hence Ast req=	720	mm^2
A_{st} provided =	753.9822	mm^2 >Ast reqd, Hence OK

Hence provide 12 mm dia steel at 150c/c top and bottom steel

Check for uplift :-

Uplift load =	150	kN	Refer reaction sheet for footing group 1(84m)
Total downward load =	∑ Fy		From L/C 206 "footing group 2 (84m) staad)
=	1129	kN	

Total uplift load < total downward load, Hence safe

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Check for Sliding :-

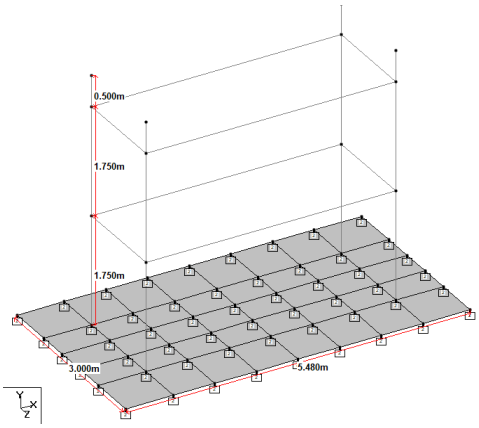
Resultant force= SRSS of Max of Fx + Fz
= 367 kN Refer reaction sheet for footing group 2(84m)

Coefficient of friction used = 0.57

Total downward force = ∑ Fy From L/C 211 "footing group 2 (84m) staad)
= 1415 kN

= 806.55 kN Hence safe

For footing group 3 (99m)



Calculation of steel required for footing raft :-

Thickness of footing =	600	mm	
Clear cover =	50	mm	
Dia of bar =	12	mm	
Spacing of bar =	120	mm	
Grade of concrete =	M	30	
Grade of steel =	Fe	500	
Bending moment=	205	kN-m	Max Mx+Mxy (From staad model)
Effective depth =	556	mm	

$$A_s = \frac{0.5\, f_c}{f_y}\left(1 - \sqrt{1 - \frac{4.6\, M_u}{f_c\, b\, d^2}}\right) bd$$

A_{st} = 870.7495 mm^2

$A_{st\,min}$ = 720 mm^2

Hence Ast req= 870.7495 mm^2

A_{st} provided = 942.4778 mm^2 >Ast reqd, Hence OK

Hence provide 12 mm dia steel at 120c/c top and bottom steel

Check for uplift :-

Uplift load = 801 kN — Refer reaction sheet for footing group 3(99m)

Total downward load = ∑ Fy — From L/C 204 "footing group 3 (99m) staad)

= 1151 kN

Total uplift load < total downward load, Hence safe

Check for Sliding :-

Resultant force= SRSS of Max of Fx + Fz

= 324 kN — Refer reaction sheet for footing group 3(99m)

Coefficient of friction used = 0.57

Total downward force = ∑ Fy — From L/C 211 "footing group 3 (99m) staad)

= 2798 kN

= 1594.86 kN Hence safe

Check for Overturning :-

Consider load combination 219. Maximum moment on the footing due to wind load in Z direction is 168kN-m

Hence, P/A±M/Z =

P = 2798kN (Summation staad reactions for 219L/C)

A = 16.44

M= 168kN-m (From Staad reaction for L/C219)

Z = 8.22 m3

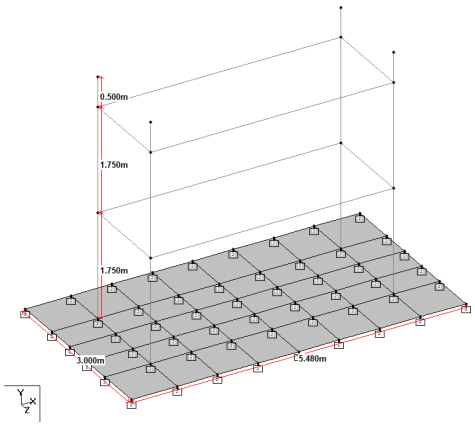
Hence, P/A+M/Z = 190.6326 kN/m2 < SBC 13x1.25 T/m2

Hence, P/A-M/Z = 149.7567 kN/m2 > 0T/m2

As both reactions are positive overturning wont happen

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For footing group 4 (99m)



Calculation of steel required for footing raft :-

Thickness of footing =	600	mm	
Clear cover =	50	mm	
Dia of bar =	12	mm	
Spacing of bar =	150	mm	
Grade of concrete =	M	30	
Grade of steel =	Fe	500	
Bending moment=	170	kN-m	Max Mx+Mxy (From staad model)
Effective depth =	556	mm	

$$A_s = \frac{0.5\, f_c}{f_y}\left(1 - \sqrt{1 - \frac{4.6\, M_u}{f_c b\, d^2}}\right) bd$$

A_{st} =	718.7219	mm^2	
$A_{st\,min}$ =	720	mm^2	
Hence Ast req=	720	mm^2	
A_{st} provided =	753.9822	mm^2	>Ast reqd, Hence OK

Hence provide 12 mm dia steel at 150c/c top and bottom steel

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Check for uplift :-

Uplift load = 581 kN — Refer reaction sheet for footing group 4(99m)

Total downward load = ∑ Fy — From L/C 206 "footing group 4 (99m) staad)

= 1277 kN

Total uplift load < total downward load, Hence safe

Check for Sliding :-

Resultant force= SRSS of Max of Fx + Fz

= 114 kN — Refer reaction sheet for footing group 4(99m)

Coefficient of friction used = 0.57

Total downward force = ∑ Fy — From L/C 211 "footing group 4 (99m) staad)

= 2467 kN

= 1406.19 kN Hence safe

ANNEXURE A – STAAD OUTPUT